Introduction to Physics-Inspired Energy Transition Neural Network (PETNN)

Ye Kyaw Thu,
LU Lab., Myanmar,
LU Talk, 25 May 2025

Overview

- Literature Review
- RNN, LSTM & GRU
- PETNN
- Experimental Setting
- Implementation
- SOTA Results
- Conclusion

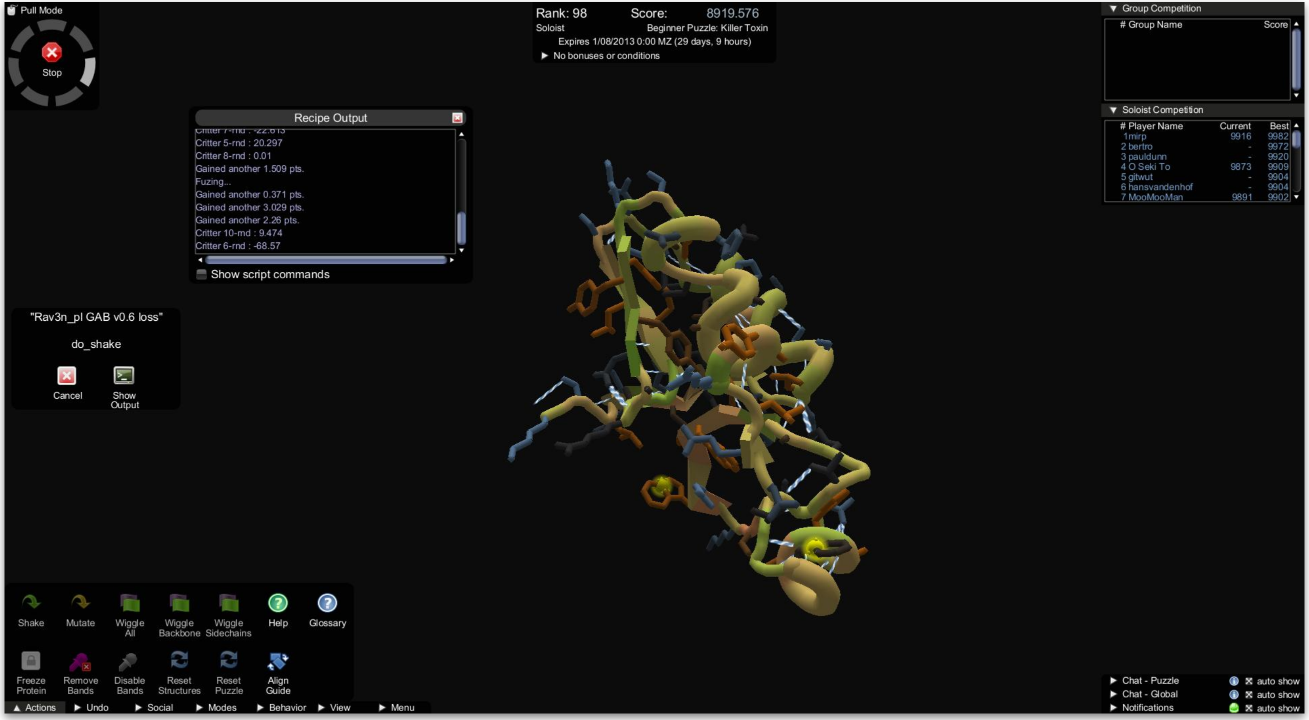
Literature Review

- Machine learning & Physics
  - Physics Informed Neural Networks (PINNs)
  - Physics Consistent Neural Networks (PCNNs)

Machine Learning & Physics

- What can ML do for Physics?
- Protein folding problem (i.e. BioPhysics, BioChemistry)
- It is the challenge of predicting the three-dimensional structure of a protein from its amino acid sequence
- Can We AlphaFold Our Way Out of the Next Pandemic? (Matthew K. Higgins, 2021)
- CASP competition and DeepMind
- FoldIt Game

Machine Learning & Physics (ML4Physics)

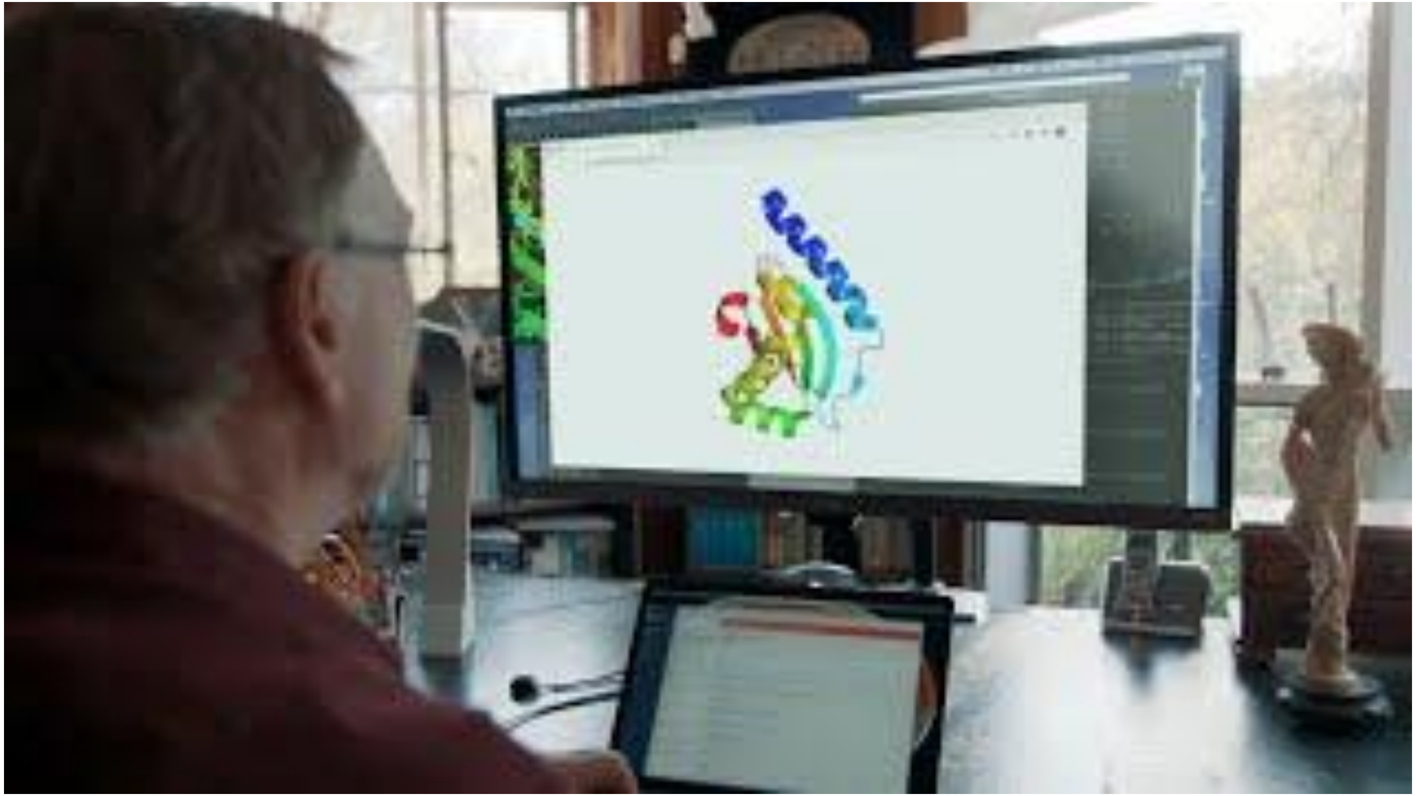


- Foldit is a revolutionary scientific discovery game that allows players to contribute to biochemistry by folding and designing proteins.

Machine Learning & Physics (ML4Physics)

YouTube Link: https://www.youtube.com/watch?v=UIdbuI4fiZs

Machine Learning & Physics (ML4Physics)



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Physics Informed Neural Networks (PINN)

Journal of Computational Physics 378 (2019) 686–707

Contents lists available at ScienceDirect

Journal of Computational Physics

www.elsevier.com/locate/jcp

Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations

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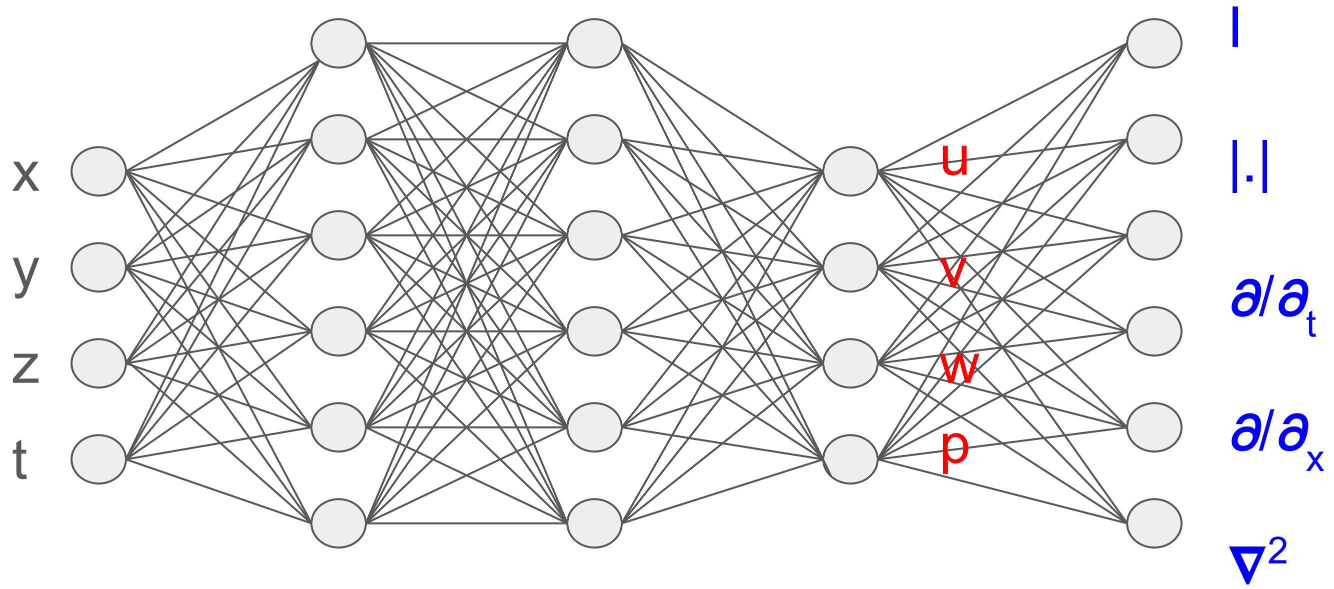
Physics Informed Neural Networks (PINN)



$$\sum\nolimits_{\text{Data}} \|\hat{\mathbf{u}}(\mathbf{x}_j) - \mathbf{u}(\mathbf{x}_j)\|_2^2 \;\; \sum\nolimits_{\text{Virtual}} \|\mathcal{N}(\mathbf{u}(\mathbf{x}_j), \mathbf{x}_j, t)\|_2^2$$

Physics Informed Neural Networks (PINN)

$$\sum_{\text{Data}} \|\hat{\mathbf{u}}(\mathbf{x}_j) - \mathbf{u}(\mathbf{x}_j)\|_2^2$$

- **Explanation:** This term represents the loss associated with the data or observed values. Here, $\hat{\mathbf{u}}(\mathbf{x}_j)$ is the observed or ground-truth velocity field (or solution) at spatial points $\mathbf{x}_j$, and $\mathbf{u}(\mathbf{x}_j)$ is the predicted velocity field from the neural network. The L_2 norm ($\| \cdot \|_2^2$) measures the squared difference between the predicted and actual values, and the summation over the "Data" set indicates this is computed across all data points. This term ensures the neural network's predictions align with the available training data.

Physics Informed Neural Networks (PINN)

$$\sum_{\text{Virtual}} \|\mathcal{N}(\mathbf{u}(\mathbf{x}_j), \mathbf{x}_j, t)\|_2^2$$

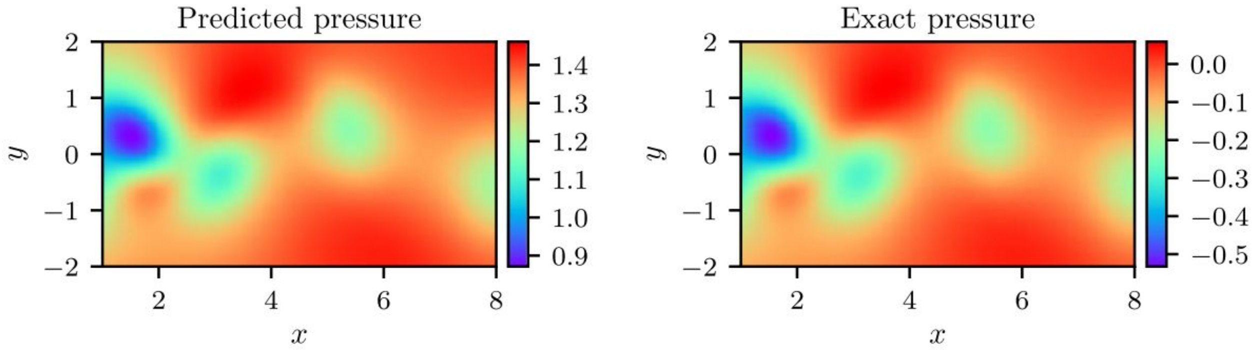
- **Explanation:** This term enforces the physical laws encoded by the PDE. $\mathcal{N}(\mathbf{u}(\mathbf{x}_j), \mathbf{x}_j, t)$ is the residual of the PDE, where $\mathbf{u}$ is the predicted solution, $\mathbf{x}_j$ are spatial coordinates, and t is time. The L_2 norm of this residual, summed over the "Virtual" set (typically a set of collocation points), quantifies how well the neural network's output satisfies the PDE. The "Virtual" points are often randomly sampled within the domain to enforce the equation everywhere, not just at data points.

Incompressible Navier-Stokes equations:

$$\nabla \cdot \mathbf{u} = 0$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \frac{1}{\mathrm{Re}}\nabla^2 \mathbf{u}$$

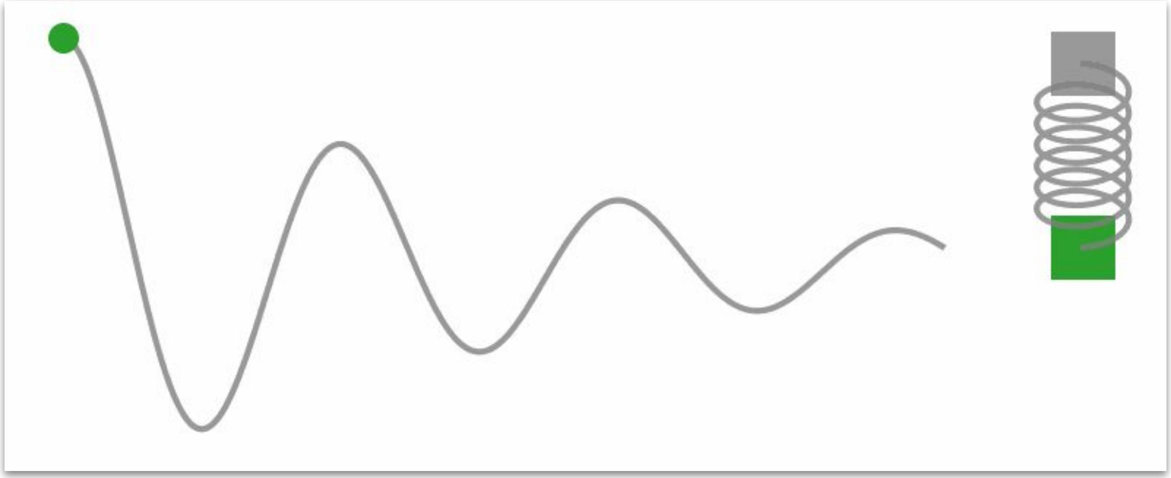
Physics Informed Neural Networks (PINN)



Correct PDE	$u_t + (uu_x + vu_y) = -p_x + 0.01(u_{xx} + u_{yy})$ $v_t + (uv_x + vv_y) = -p_y + 0.01(v_{xx} + v_{yy})$
Identified PDE (clean data)	$u_t + 0.999(uu_x + vu_y) = -p_x + 0.01047(u_{xx} + u_{yy})$ $v_t + 0.999(uv_x + vv_y) = -p_y + 0.01047(v_{xx} + v_{yy})$
Identified PDE (1% noise)	$u_t + 0.998(uu_x + vu_y) = -p_x + 0.01057(u_{xx} + u_{yy})$ $v_t + 0.998(uv_x + vv_y) = -p_y + 0.01057(v_{xx} + v_{yy})$

Fig. 4. *Navier–Stokes equation: Top:* Predicted versus exact instantaneous pressure field $p(t, x, y)$ at a representative time instant. By definition, the pressure can be recovered up to a constant, hence justifying the different magnitude between the two plots. This remarkable qualitative agreement highlights the ability of *physics-informed neural networks* to identify the entire pressure field, despite the fact that no data on the pressure are used during model training. *Bottom:* Correct partial differential equation along with the identified one obtained by learning λ_1, λ_2 and $p(t, x, y)$.

Physics Informed Neural Networks (PINN)



- With PINNs, we can solve problems related to the damped harmonic oscillator. If you're interested, check out this Jupyter Notebook: https://github.com/benmoseley/DLSC-2023/blob/main/lecture-5/PINN%20demo.ipynb

Physics Consistent Neural Networks (PCNNs)

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

Physically Consistent Neural Networks for building thermal modeling: Theory and analysis

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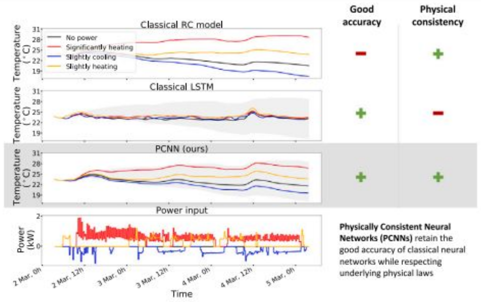
GRAPHICAL ABSTRACT



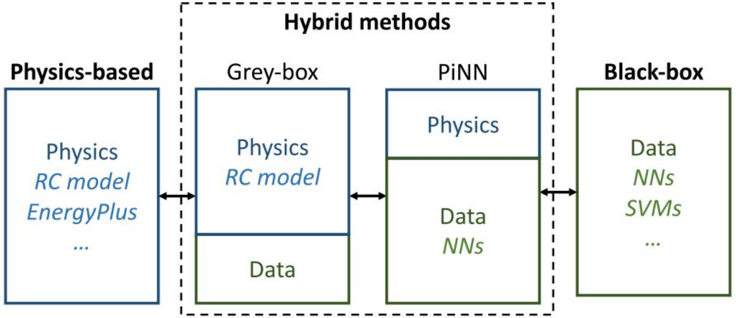
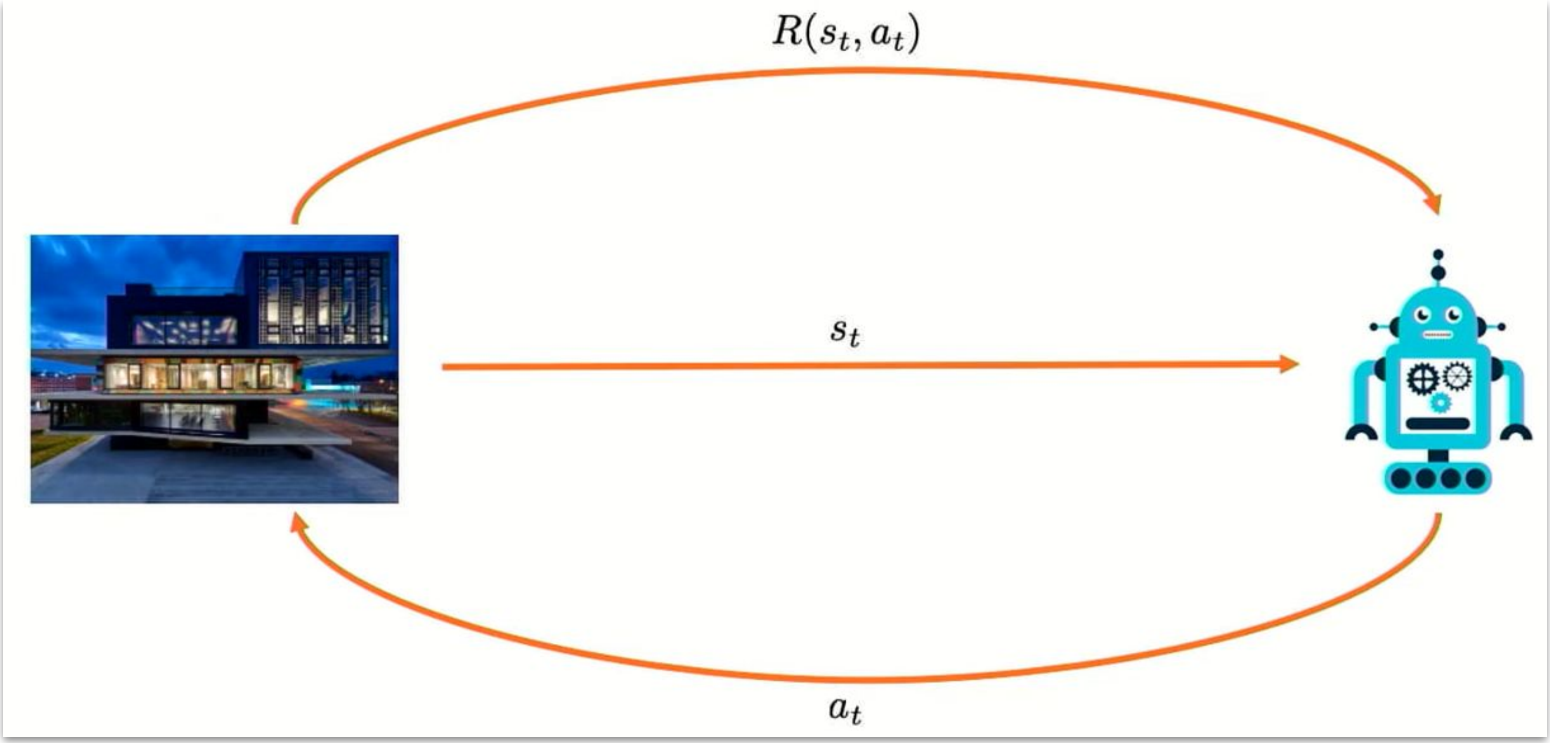


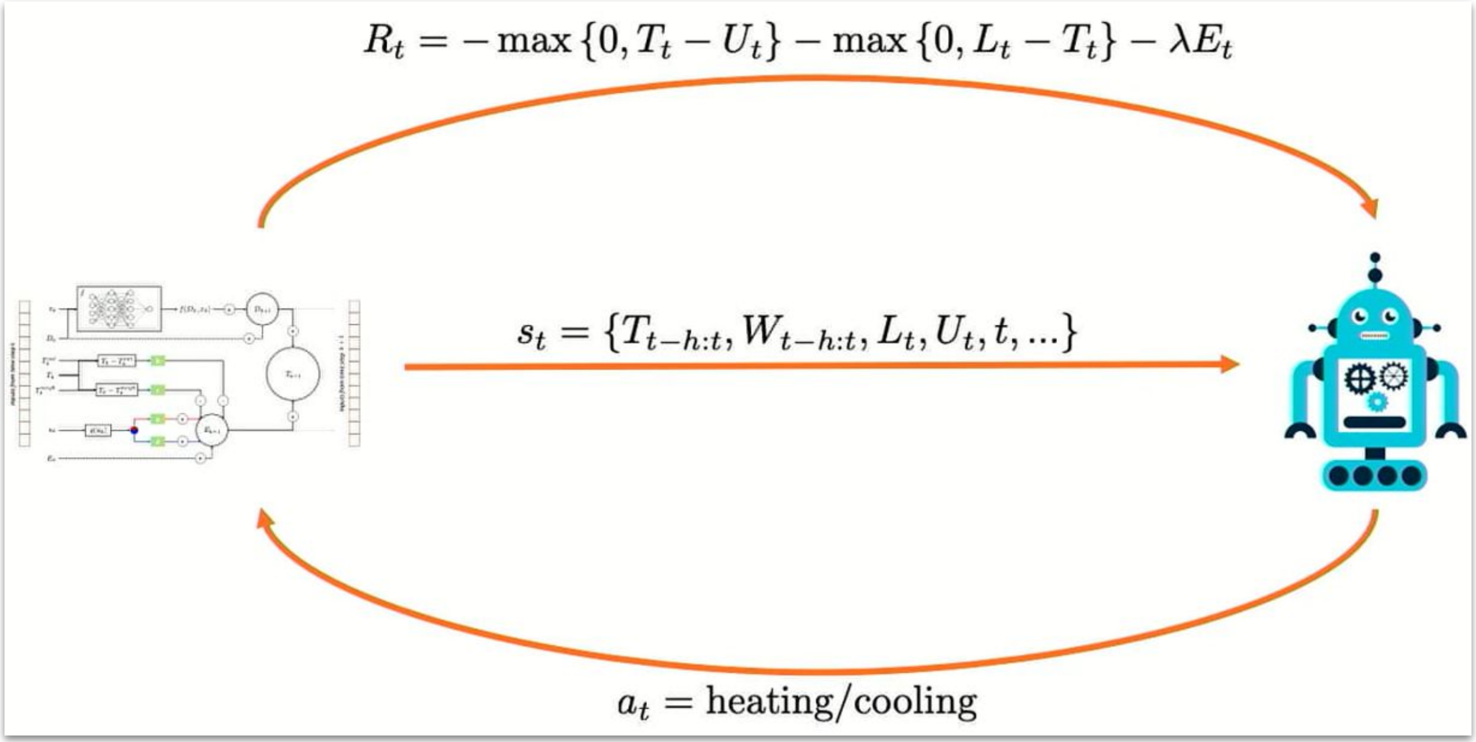
Fig. 2. Structural differences between the different methods..

Physics Consistent Neural Networks (PCNNs)



Source: https://www.youtube.com/watch?v=HCO8qmsvcgo&t=1597s

Physics Consistent Neural Networks (PCNNs)



Source: https://www.youtube.com/watch?v=HCO8qmsvcgo&t=1597s

Physics Consistent Neural Networks (PCNNs)

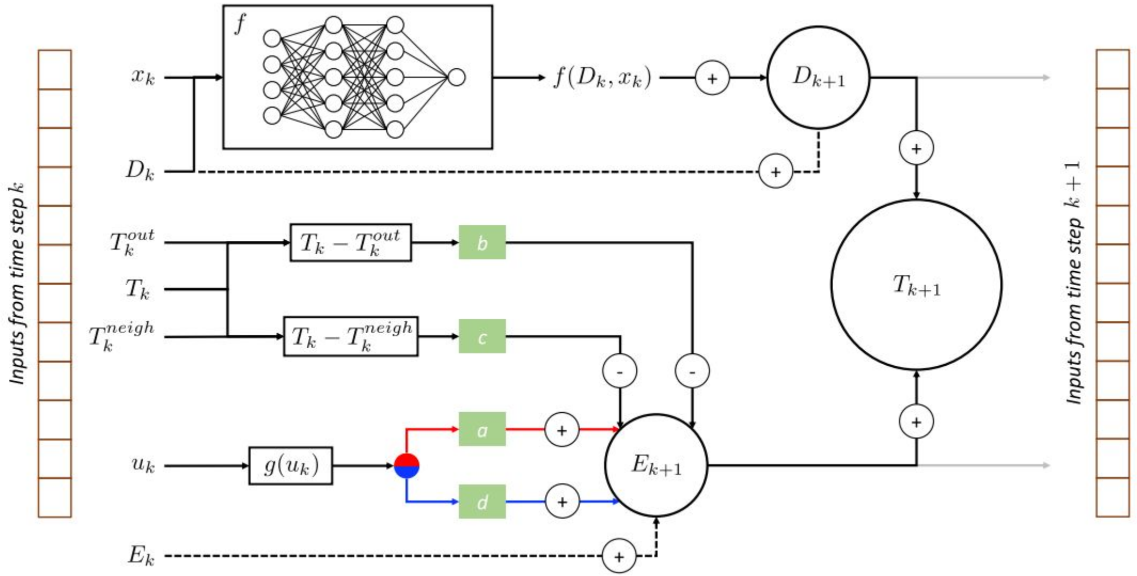


Fig. 3. The proposed PCNN architecture used recursively at each time step. The control inputs u, transformed into power inputs by the function g, and the losses to the environment $b(T - T^{out})$ and neighboring zone $c(T - T^{neigh})$ all influence an energy accumulator E, which accumulates or dissipates energy at each time step. Here, the separation between red and blue lines signals a different treatment of the power inputs in the heating and cooling case, respectively, since they are scaled by different constants a and d. The accumulated energy is then added to the unforced dynamics D, modeled by a residual NN that takes all the features apart from u, T^{out}, and T^{neigh} – gathered in x – as input, to get the final zone temperature prediction T..

Source:
https://www.sciencedirect.com/science/article/pii/S0306261922010819?ref=pdf_download&fr=RR-2&rr=945c7e0cca0e893d

Physics Consistent Neural Networks (PCNNs)

- The eye reveals both share similar physical behavior
- PCNNs adhere to physical laws despite using NNs
- PCNNs outperform RC models in data fitting and accuracy

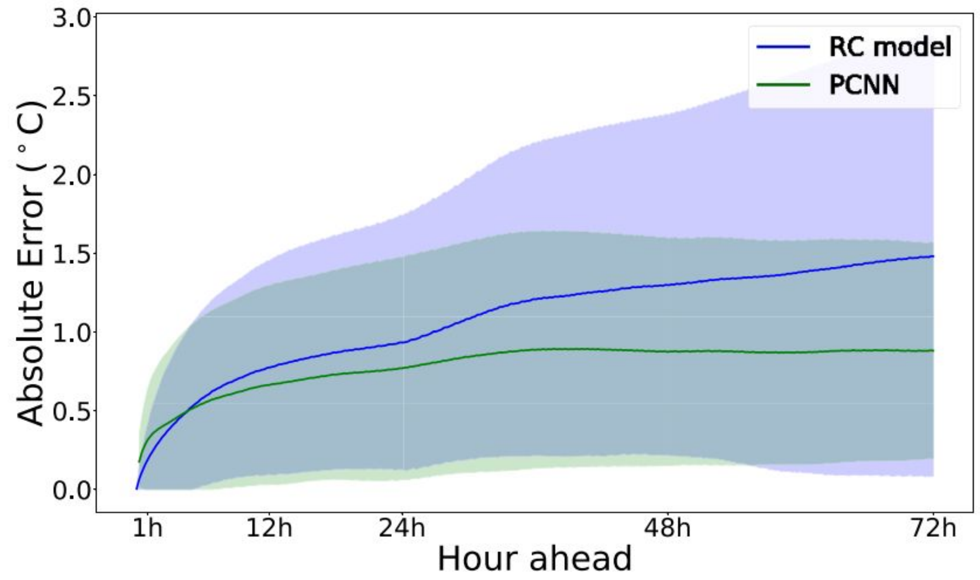


Fig. 5. Mean and standard deviation of the error at each time step of the prediction horizon for both the RC model in blue and the PCNN in green, where the statistics were computed from almost 2000 predictions from the validation set..

Source: https://www.sciencedirect.com/science/article/pii/S0306261922010819?ref=pdf_download&fr=RR-2&rr=945c7e0cca0e893d

Physics Consistent Neural Networks (PCNNs)

- DRL agents align with top-performing paths (determined afterward)
- The authors verified this using statistical tests on 2000 three-day runs

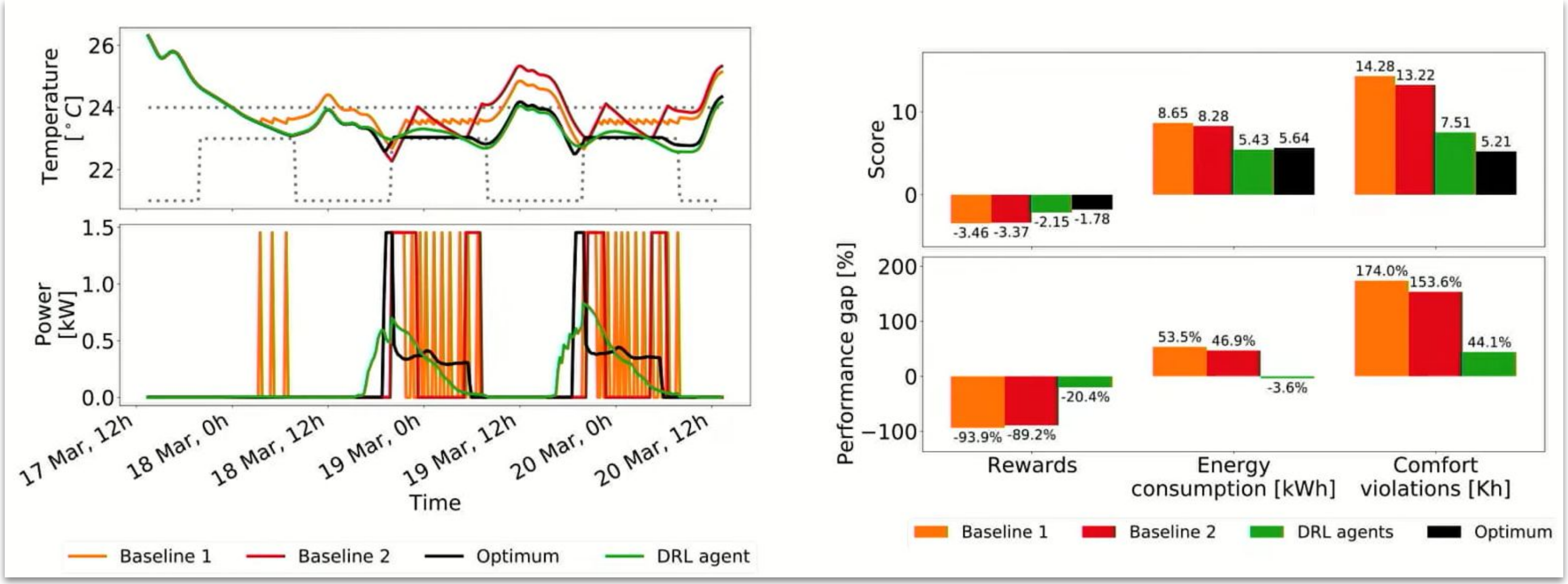


Fig. Simulation results

Physics-guided neural networks (phygnn)

github.com/NREL/phygnn

README   BSD-3-Clause license

phygnn

Documentation passing | pytests_tf_latest passing | Lint Code Base passing | python 3.8 | 3.9 | 3.10 | 3.11
pypi package 0.0.31 | Anaconda.org 0.0.31 | license BSD_3_Clause | codecov 86% | DOI 10.5281/zenodo.15277734
launch binder

phygnn (fi-geon | ˈfi-jən) noun.

> 1. a physics-guided neural network
> 2. a rare and mythical bird

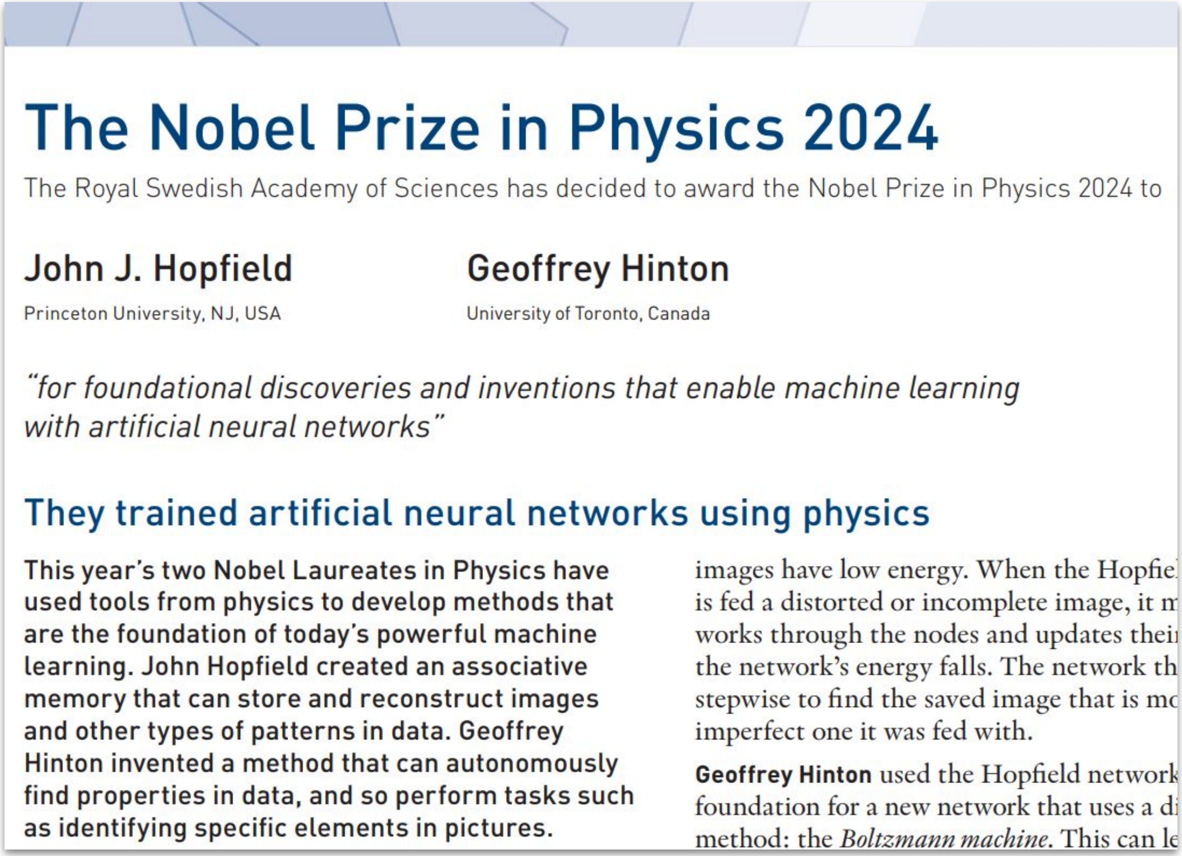
This implementation of physics-guided neural networks augments a traditional neural network loss function with a generic loss term that can be used to guide the neural network to learn physical or theoretical constraints. phygnn enables scientific software developers and data scientists to easily integrate machine learning models into physics and engineering applications. This framework should help alleviate some challenges that are often encountered when applying purely data-driven machine learning models to scientific applications, such as when machine learning models produce physically inconsistent results or have trouble generalizing to out-of-sample scenarios.

For details on the phygnn class framework see the phygnn class documentation here.

Machine Learning & Physics

- What can Physics do for ML?
- The link between statistical mechanics and learning theory emerged in the mid-1980s when example-based statistical learning replaced logic and rule-based AI.
- Valiant’s learnable theory (Valiant, 1984).
- Hopfield’s associative memory model (Hopfield, 1982), igniting the use of spin glass concepts in neural networks.
- This was highlighted by Amit, Gutfreund, and Sompolinsky’s memory capacity analysis of the Hopfield model (Amit et al., 1985) and subsequent studies.
- And more.

Machine Learning & Physics



The Nobel Prize in Physics 2024

The Royal Swedish Academy of Sciences has decided to award the Nobel Prize in Physics 2024 to

John J. Hopfield
Princeton University, NJ, USA

Geoffrey Hinton
University of Toronto, Canada

"for foundational discoveries and inventions that enable machine learning with artificial neural networks"

They trained artificial neural networks using physics

This year's two Nobel Laureates in Physics have used tools from physics to develop methods that are the foundation of today's powerful machine learning. John Hopfield created an associative memory that can store and reconstruct images and other types of patterns in data. Geoffrey Hinton invented a method that can autonomously find properties in data, and so perform tasks such as identifying specific elements in pictures.

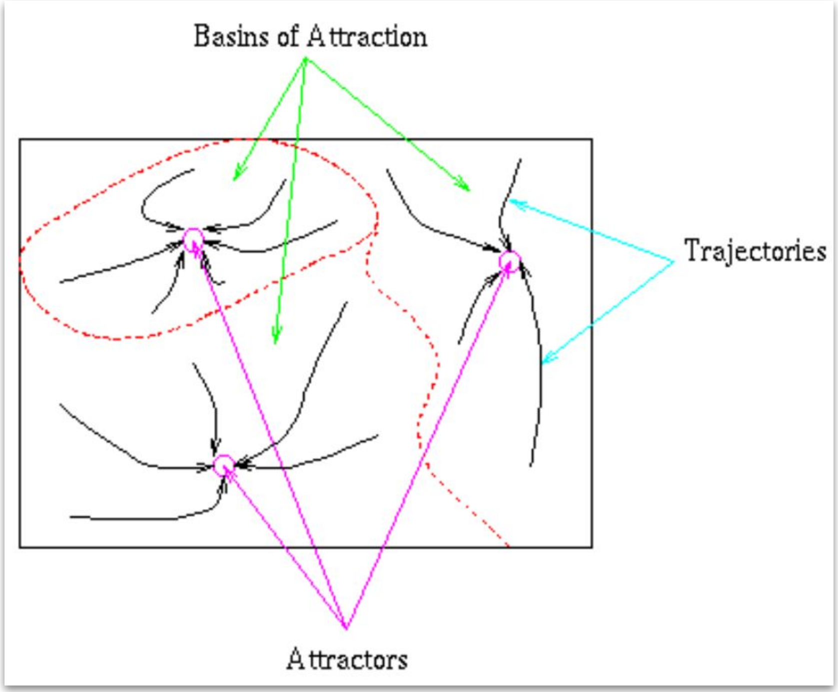
images have low energy. When the Hopfie
is fed a distorted or incomplete image, it m
works through the nodes and updates thei
the network's energy falls. The network th
stepwise to find the saved image that is mo
imperfect one it was fed with.

Geoffrey Hinton used the Hopfield network
foundation for a new network that uses a di
method: the *Boltzmann machine*. This can le

Source:
https://www.nobelprize.org/uploads/2024/11/press-physicsprize2024-2.pdf

Prize Fig:
https://www.nobelprize.org/prizes/

Machine Learning & Physics (The Hopfield Network)



- Figure shows a schematic of our plastic sheet. The low points are the attractors. The region within which the drop will fall to a given attractor is its basin of attraction and the paths taken by the drop are called trajectories.

Fig. Attractors, basins of attraction and trajectories.
Source: https://staff.itee.uq.edu.au/janetw/cmc/chapters/Hopfield/

Machine Learning & Physics (The Hopfield Network)

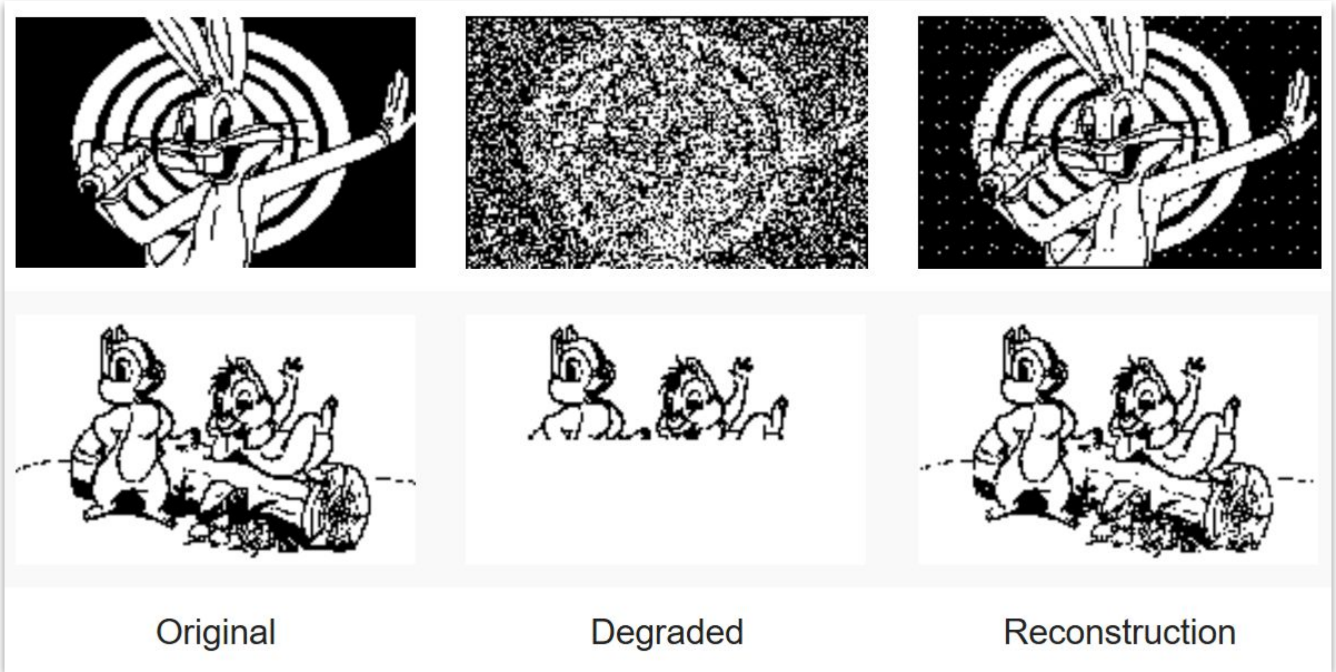
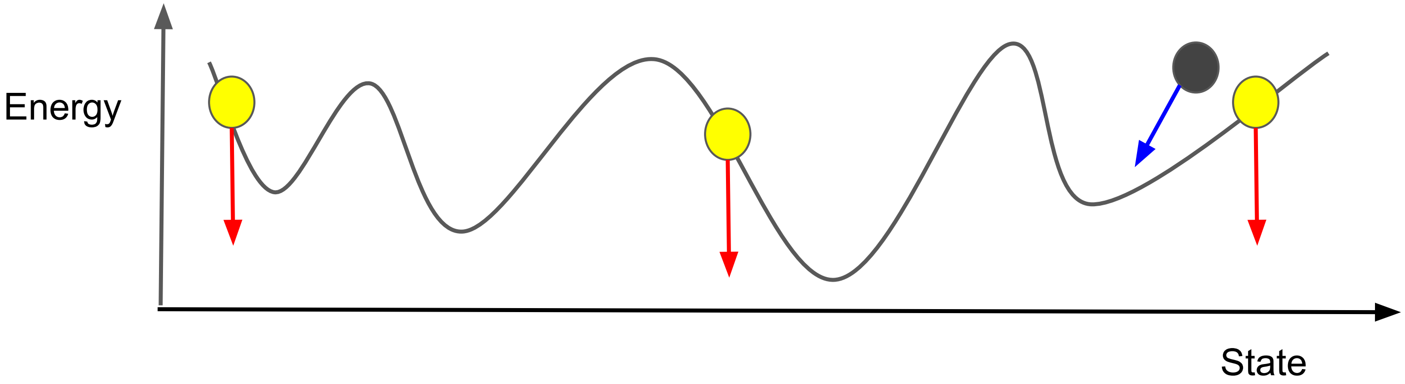


Fig. Hopfield network reconstructing degraded images from noisy (top) or partial (bottom) cues. Source: https://staff.itee.uq.edu.au/janetw/cmc/chapters/Hopfield/

Machine Learning & Physics (The Hopfield Network)

$$\mathrm{W} = \mathrm{W} + \eta \left(\sum_{\mathbf{y} \in Y_p} \mathbf{y}\mathbf{y}^T - \sum_{\mathbf{y} \in Y_p \text{ and } \mathbf{y}=\text{valley}} \mathbf{y}\mathbf{y}^T \right)$$



- Lower energy at valid memories

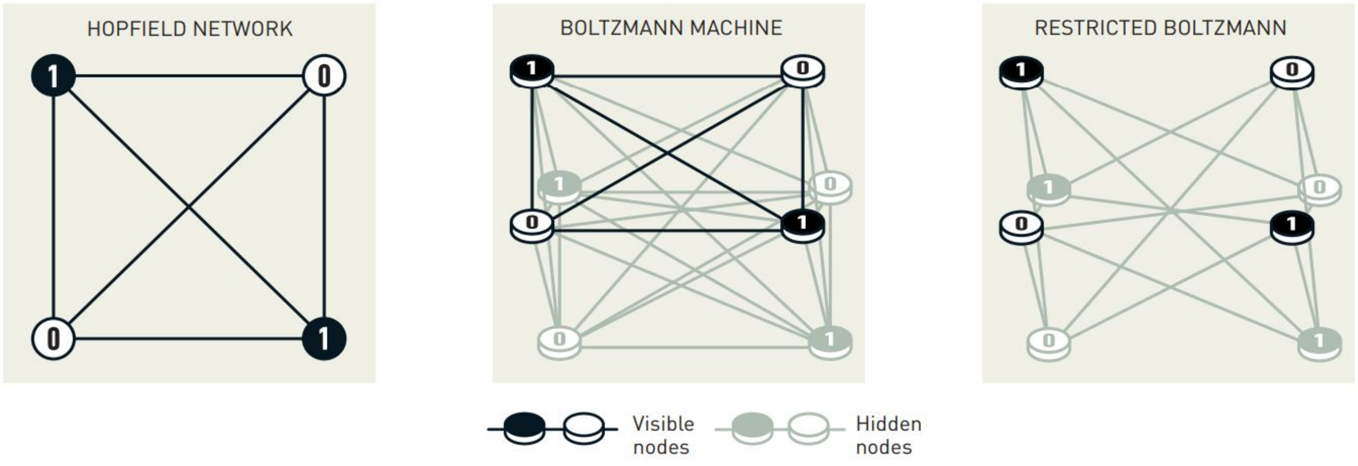
Fig. Initialize the network at valid memories.
Source: https://deeplearning.cs.cmu.edu/F20/document/slides/lecx3.boltzmann.pdf

Machine Learning & Physics (The Boltzmann Machine)

Geoffrey Hinton used the Hopfield network as the foundation for a new network that uses a different method: the *Boltzmann machine*. This can learn to recognise characteristic elements in a given type of data. Hinton used tools from statistical physics, the science of systems built from many similar components. The machine is trained by feeding it examples that are very likely to arise when the machine is run. The Boltzmann machine can be used to classify images or create new examples of the type of pattern on which it was trained. Hinton has built upon this work, helping initiate the current explosive development of machine learning.

Source: https://www.nobelprize.org/uploads/2024/11/press-physicsprize2024-2.pdf

Machine Learning & Physics (The Boltzmann Machine)



John Hopfield's associative memory is built so that all the nodes are connected to each other. Information is fed in and read out from all the nodes.

Geoffrey Hinton's Boltzmann machine is often constructed in two layers, where information is fed in and read out using a layer of *visible* nodes. They are connected to *hidden* nodes, which affect how the network functions in its entirety.

In a restricted Boltzmann machine, there are no connections between nodes in the same layer. The machines are frequently used in a chain, one after the other. After training the first restricted Boltzmann machine, the content of the hidden nodes is used to train the next machine, and so on.

Source: https://www.nobelprize.org/uploads/2024/10/popular-physicsprize2024-2.pdf

Machine Learning & Physics

arXiv > physics > arXiv:1903.10563

Search...

Help

Physics > Computational Physics

[Submitted on 25 Mar 2019 (v1), last revised 6 Dec 2019 (this version, v2)]

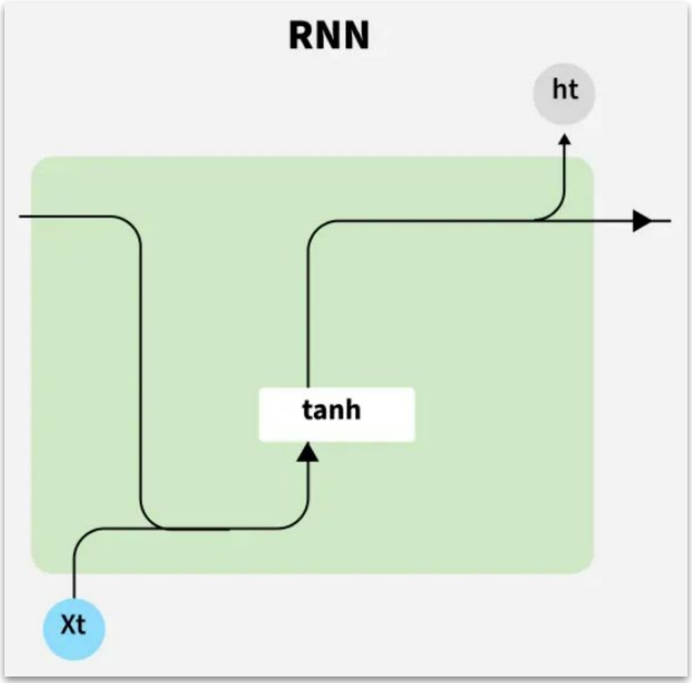
Machine learning and the physical sciences

C[x] CODE Notebook

Giuseppe Carleo, Ignacio Cirac, Kyle Cranmer, Laurent Daudet, Maria Schuld, Naftali Tishby, Leslie Vogt-Maranto, Lenka Zdeborová

Machine learning encompasses a broad range of algorithms and modeling tools used for a vast array of data processing tasks, which has entered most scientific disciplines in recent years. We review in a selective way the recent research on the interface between machine learning and physical sciences. This includes conceptual developments in machine learning (ML) motivated by physical insights, applications of machine learning techniques to several domains in physics, and cross-fertilization between the two fields. After giving basic notion of machine learning methods and principles, we describe examples of how statistical physics is used to understand methods in ML. We then move to describe applications of ML methods in particle physics and cosmology, quantum many body physics, quantum computing, and chemical and material physics. We also highlight research and development into novel computing architectures aimed at accelerating ML. In each of the sections we describe recent successes as well as domain-specific methodology and challenges.

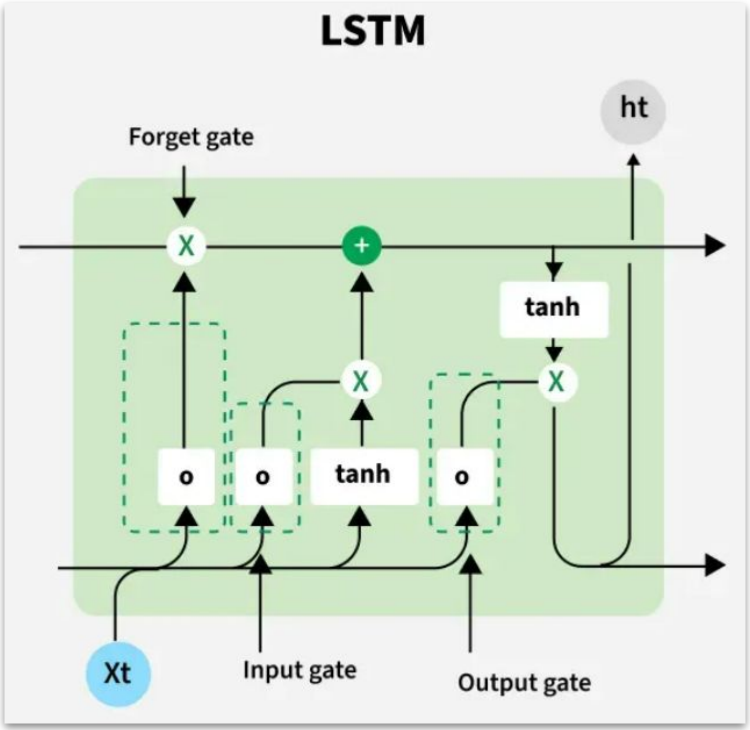
RNN, LSTM and GRU



- Designed for sequential data with loops to retain past info.
- Architecture: Input → Hidden (recurrent) → Output.
- Great for short-term dependencies (e.g., language modeling).
- Struggles with long-term dependencies due to vanishing gradients.
- Limits performance in tasks like translation or speech recognition.

Source: https://www.geeksforgeeks.org/rnn-vs-lstm-vs-gru-vs-transformers/

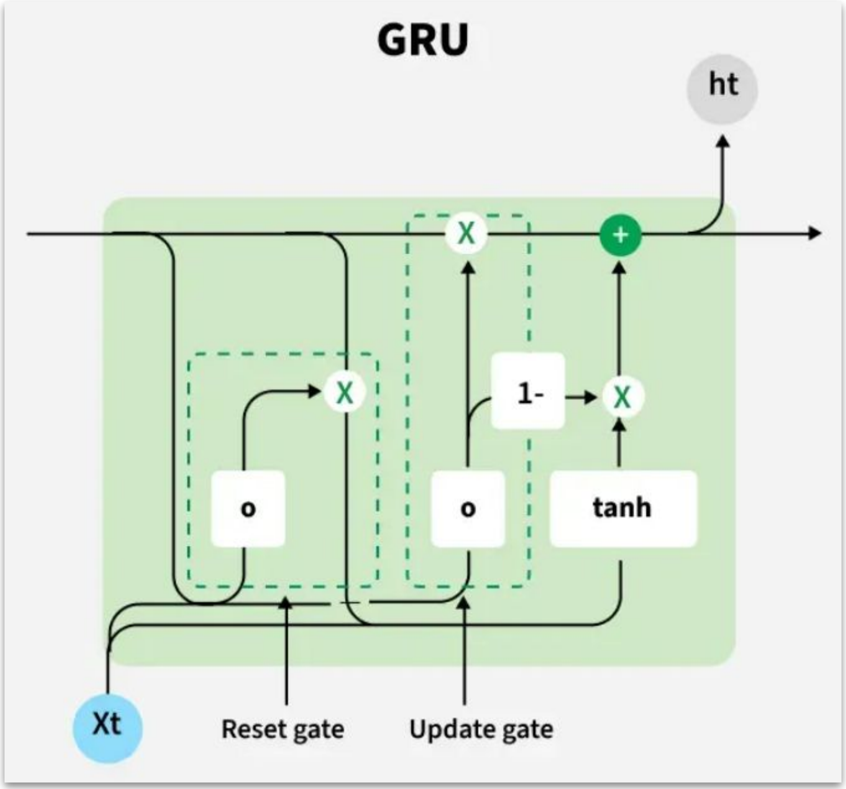
RNN, LSTM and GRU



- Advanced RNN variant to tackle vanishing gradient issues.
- Uses memory cells with gates: Input (store), Forget (discard), Output (send).
- Excels at long-term dependencies (e.g., sentiment analysis, translation).
- Limitations: Computationally expensive, slow training, struggles with very long sequences, limited parallelization.

Source: https://www.geeksforgeeks.org/rnn-vs-lstm-vs-gru-vs-transformers/

RNN, LSTM and GRU



- Simplified LSTM with a single update gate (combines input/forget) and reset gate.
- Update Gate: Retains key past info; Reset Gate: Discards irrelevant data.
- Balances efficiency and performance, often outperforming LSTMs.
- Ideal for resource-constrained tasks with good accuracy.

Source: https://www.geeksforgeeks.org/rnn-vs-lstm-vs-gru-vs-transformers/

Physics-inspired Energy Transition Neural Network (PETNN)

arXiv > cs > arXiv:2505.03281

Search
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Computer Science > Machine Learning

[Submitted on 6 May 2025]

Physics-inspired Energy Transition Neural Network for Sequence Learning

Zhou Wu, Junyi An, Baile Xu, Furao Shen, Jian Zhao

Recently, the superior performance of Transformers has made them a more robust and scalable solution for sequence modeling than traditional recurrent neural networks (RNNs). However, the effectiveness of Transformer in capturing long-term dependencies is primarily attributed to their comprehensive pair-modeling process rather than inherent inductive biases toward sequence semantics. In this study, we explore the capabilities of pure RNNs and reassess their long-term learning mechanisms. Inspired by the physics energy transition models that track energy changes over time, we propose a effective recurrent structure called the``Physics-inspired Energy Transition Neural Network" (PETNN). We demonstrate that PETNN's memory mechanism effectively stores information over long-term dependencies. Experimental results indicate that PETNN outperforms transformer-based methods across various sequence tasks. Furthermore, owing to its recurrent nature, PETNN exhibits significantly lower complexity. Our study presents an optimal foundational recurrent architecture and highlights the potential for developing effective recurrent neural networks in fields currently dominated by Transformer.

PETNN

which is common in RNN-based models, and incorporating memory mechanisms, as seen in models like LSTM and GRU. While RNN-based models, due to their simpler structure, have lower time complexity, they rely heavily on memory, leading to the well-known issue of forgetting past information. LSTM and GRU address this by introducing dedicated memory and forget mechanisms, improving their ability to capture long-term dependencies. However, both models still rely on sigmoid-based gates for controlling memory, which can lead to information leakage over long sequences. This limitation makes them less effective compared to Transformer models, which capture long-range dependencies by modeling all pairwise interactions within the sequence.

In this paper, we propose **Physical-inspired Energy Transition Neural Network (PETNN)**, which is a novel recurrent structure for effective sequence modeling. Initially, we observed that the energy transition model closely resembles

the memory cell, since the energy change is a time function. This observation inspired the core idea of PETNN, i.e., using energy as the cell state for updates. To address this, we propose a novel memory mechanism in PETNN. Unlike LSTM, which relies on predefined memory and forget gates, PETNN empowers the neurons to autonomously determine what information to learn and update based on the energy state, a process we refer to as the *self-selective information mixing method*. This approach offers two key advantages: first, it allows neurons to dynamically control the proportion of information updates; second, it enables neurons to decide the storage duration of relevant information.

PETNN (3 main contributions)

networks to solve non-sequential problems, laying a solid foundation for our future work. Our contributions are as follows:

- We propose PETNN, a recurrent neural networks that incorporates wisdom from physics-based spatiotemporal model (energy transition model).
- We analyze PETNN's structure and demonstrate its advantages in handling long-term dependencies.
- Results show that PETNN outperforms Transformers on several sequence tasks. Ablation experiments confirm the effectiveness of PETNN's memory mechanism, providing new insights for sequence learning research.

PETNN

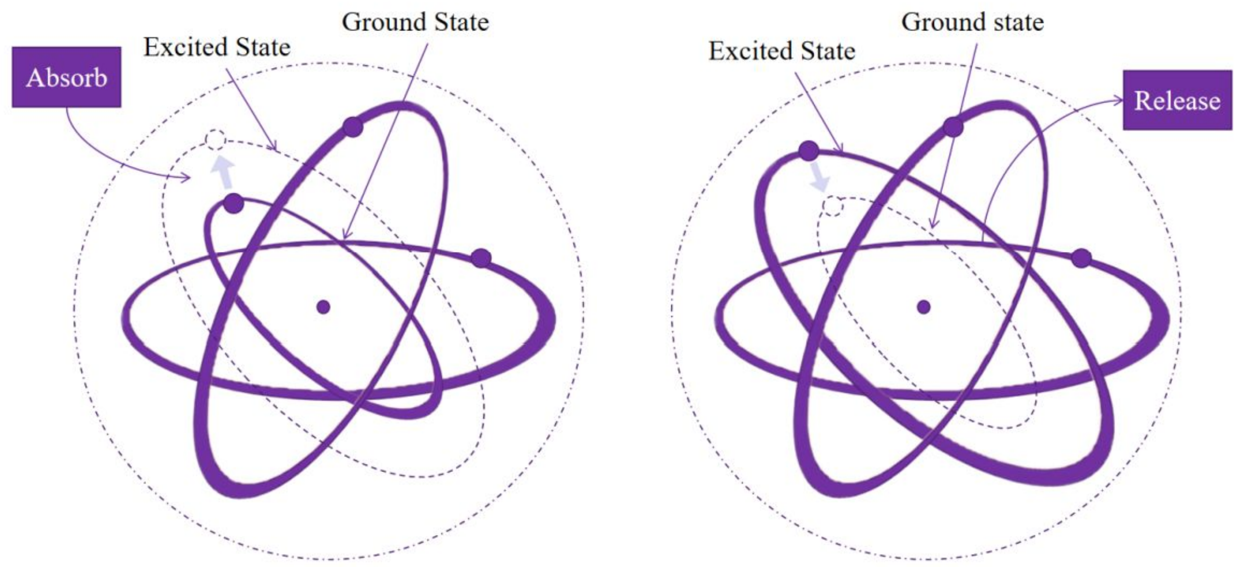

Figure 1. Quantum theory explains the energy transition model through the processes of absorption and release.

Source: https://arxiv.org/pdf/2505.03281

PETNN vs ST-PCNN

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RESEARCH ARTICLE

Physics-Coupled Spatio-Temporal Active Learning for Dynamical Systems

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PETNN vs ST-PCNN

ABSTRACT Spatio-temporal forecasting is of great importance in a wide range of dynamic systems applications, such as earth science, transport planning, etc. These applications rely on accurate predictions of spatio-temporal structured data reflecting real-world phenomena. A stunning characteristic is that the dynamical system is not only driven by some physics laws but also impacted by the localized factor in spatial and temporal regions. One of the major challenges is to infer the underlying causes, which generate the perceived data stream and propagate the involved causal dynamics through the distributed observing units. Another challenge is that the success of machine learning-based predictive models requires massive annotated data for model training. However, the acquisition of high-quality annotated data is objectively manual and tedious as it needs a considerable amount of human intervention, making it infeasible in fields that require high levels of expertise. To tackle these challenges, we advocate a spatio-temporal physics-coupled neural networks (ST-PCNN) model to learn the underlying physics of the dynamical system and further couple the learned physics to assist the learning of the recurring dynamics. To deal with data-acquisition constraints, an active learning mechanism with Kriging for actively acquiring the most informative data is proposed for ST-PCNN training in a partially observable environment. Our experiments on both synthetic and real-world datasets exhibit that the proposed ST-PCNN with active learning converges to near-optimal accuracy with substantially fewer instances.

INDEX TERMS Spatio-temporal modeling, physics-informed neural networks, active learning, Gaussian process model, dynamical systems.

PETNN vs ST-PCNN

Overview of Each Proposal

Aspect	Paper 1	Paper 2
Title	*Physics-Coupled Spatio-Temporal Active Learning for Dynamical Systems*	*Physics-inspired Energy Transition Neural Network for Sequence Learning*
Core Model	ST-PCNN (Spatio-Temporal Physics-Coupled Neural Network)	PETNN (Physics-inspired Energy Transition Neural Network)
Key Idea	Combines physical laws (via PDEs) and active learning with a spatiotemporal neural network for dynamic system prediction under data constraints.	Designs a novel RNN-based structure that models memory and sequence dynamics inspired by energy transitions in physics (quantum systems).
Physical Inspiration	Uses Gaussian processes to derive PDE coefficients from partial observations, modeling homogeneity and heterogeneity in space-time data.	Mimics quantum energy transitions (excitation/decay) to design dynamic memory states and adaptive update rules.

PETNN vs ST-PCNN

🧪 Application Domains

Aspect	ST-PCNN	PETNN
Primary Application	Environmental sensing, oceanography, atmospheric systems — where physical laws and sparse data dominate	Generic sequence learning: time series forecasting, text sentiment analysis, even image classification
Dataset Types	Real-world spatiotemporal datasets (e.g., ocean sensors, weather grids)	Benchmarks like ETT, Traffic, Weather, Exchange, ACL-IMDB
Use Case Emphasis	Realistic systems constrained by sensor placement and data collection cost	Scalable sequence tasks with long-term dependencies and minimal computational burden

PETNN

Figure 2. A basic neuron in PETNN. The black waveform represents stored neuronal information over time. When the signal reaches the red dashed line, the neuron releases this information, returning to ground state.

Source: https://arxiv.org/pdf/2505.03281

PETNN

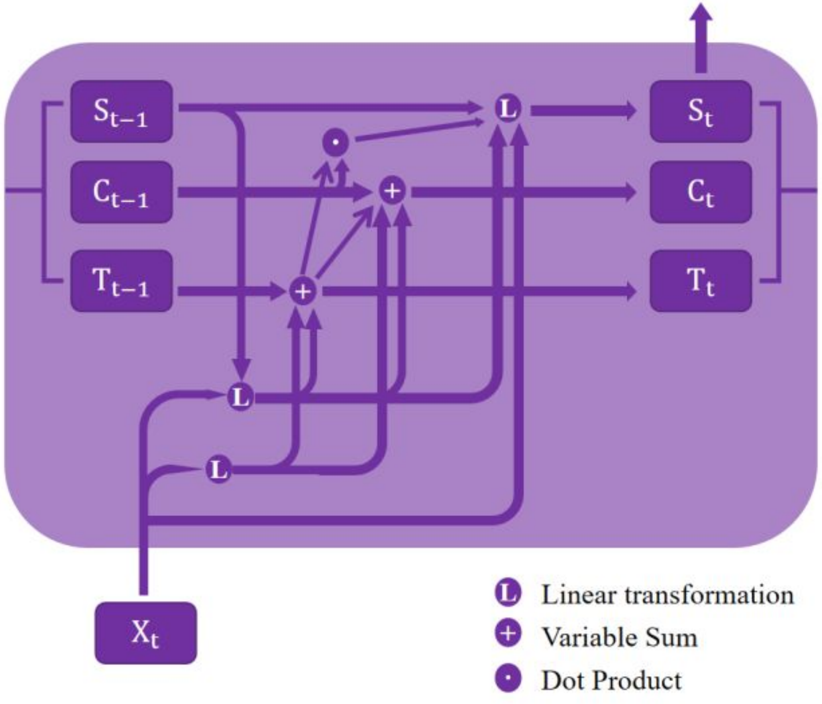


Figure 3. Internal architecture of PETNN.

3.1. Physics-inspired Energy Transition Neuron

Inspired by the energy transition model, we design a neuron structure. As shown in Figure 2, the neuron can be regarded as the atom. We use a series of variables to mathematically represent it, which are defined as follows.

- **Remaining Time $\mathbf{T}_t$**: the remaining time of cell state at step t, corresponds to the residence time of atoms in excited state.
- **Cell State $\mathbf{C}_t$**: the state of the neuron at step t, which can be regarded as the energy state of the neuron.
- **Hidden State $\mathbf{S}_t$**: the memory in time step t, as another input of the neuron at time step t+1 except input X_{t+1}.

Source: https://arxiv.org/pdf/2505.03281

PETNN

Formally, all the updating processes are

$$\begin{aligned} T_t &= R_t \cdot \sigma(T_{t-1} + Z_t) - 1 \\ C_t &= (1 - m) \cdot C_{t-1} + m \cdot I_t + Z_c \\ h_t &= f(S_{t-1}, (1 - m) \cdot C_{t-1}, X_t) \\ S_t &= \sigma\big((1 - Z_w) \cdot S_{t-1} + Z_w \cdot h_t\big) \ , \end{aligned} \tag{3}$$

where $\mathbf{h_t}$ represents the information learned at time step t, which is derived based on the state from the previous time step and the input $\mathbf{X_t}$, m is the switch of the release, f denotes the set of mapping applied to the input and previous state to compute $\mathbf{h_t}$.

PETNN (Dataset Info)

Table 7. Dataset descriptions. All the experiments use the ADAM optimizer.

Datasets	Dim	Series Length	Dataset size	Information(Frequency)
MNIST	28	{28,764}	(60000, 10000)	Handwritten digits
ACL-IMDB	300	{300,600}	(25000, 25000)	Movie Review
ETTm1,ETTm2	7	{96,192,336,720}	(34465, 11521, 11521)	Electricity(15 mins)
ETTh1,ETTh2	7	{96, 192, 336, 720}	(8545, 2881, 2881)	Electricity(15 mins)
Electricity	321	{96, 192, 336, 720}	(18317, 2633, 5261)	Electricity(Hourly)
Traffic	862	{96, 192, 336, 720}	(12185, 1757, 3509)	Transportation(Hourly)
Weather	21	{96, 192, 336, 720}	(36792, 5271, 10540)	Weather(10 mins)
Exchange	8	{96, 192, 336, 720}	(5120, 665, 1422)	Exchange rate(Daily)

PETNN (results for ACL-IMDB dataset)

Table 3. Text Sentiment Classification Task.

MODEL	PETNN	TEXTCNN	LSTM	GRU	MLP
ACC. (%)	**89**	<u>84</u>	83	81	72

PETNN (ACL-IMDB dataset)

```
emorable.(base) ye@lst-hpc3090:~/exp/petnn/classi/data/imdb/aclImdb/train/pos$ cat 9999_8.txt
The plot had some wretched, unbelievable twists. However, the chemistry between Mel Brooks and Leslie Ann Warre
n was excellent. The insight that she comes to, "There are just moments," provides a philosophical handle by wh
ich anyone could pick up, and embrace, life.<br /><br />That was one of several moments that were wonderfully m
emorable.(base) ye@lst-hpc3090:~/exp/petnn/classi/data/imdb/aclImdb/train/pos$ cat 9998_9.txt
Seeing as the vote average was pretty low, and the fact that the clerk in the video store thought it was "just
OK", I didn't have much expectations when renting this film.<br /><br />But contrary to the above, I enjoyed it
 a lot. This is a charming movie. It didn't need to grow on me, I enjoyed it from the beginning. Mel Brooks giv
es a great performance as the lead character, I think somewhat different from his usual persona in his movies.<
br /><br />There's not a lot of knockout jokes or something like that, but there are some rather hilarious scen
es, and overall this is a very enjoyable and very easy to watch film.<br /><br />Very recommended.(base) ye@lst
-hpc3090:~/exp/petnn/classi/data/imdb/aclImdb/train/pos$
```

```
(base) ye@lst-hpc3090:~/exp/petnn/classi/data/imdb/aclImdb/train/neg$ cat ./9999_3.txt
This is the kind of movie that my enemies content I watch all the time, but it's not bloody true. I only watch
it once in a while to make sure that it's as bad as I first thought it was.<br /><br />Some kind of mobsters hi
jack a Boeing 747. (That, at least, is an improvement over having Boeing hijack a good part of the Pentagon.) T
he airplane goes down in the Bermuda triangle and sinks pressurized to the bottoms, a kind of post-facto submar
ine.<br /><br />It has one of those all-star casts, the stars either falling or barely above the horizon.<br />
<br />"We're on our own!", says pilot Jack Lemon. He is so right. Except for George Kennedy. He's in all these
disaster movies.<br /><br />Watch another movie instead. Oh, not "Airport" the original. That's no good either.
 Instead, watch a decent flick about stuck airplanes like "Flight of the Phoenix."(base) ye@lst-hpc3090:~/exp/p
etnn/classi/data/imdb/aclImdb/train/neg$
```

PETNN (classification results for MNIST)

Table 8. Image classification Task on different scenarios.

MODEL	LAYER	INPUT SIZE	ACCURACY
FC	1	[1,784]	90.91%
PETNN	1	[1,784]	**96.80%**
FC	2	[1,784]	96.91%
PETNN	2	[1,784]	**97.20%**
KAN	3	[28,28]	96.70%
LSTM	1	[28,28]	98.07%
CNN	1	[28,28]	98.47%
PETNN	1	[28,28]	**99.03%**

PETNN

Table 1. Long-term forecasting tasks on basic models. The past sequence is set as 96 for all datasets. All the results are average from 4 different prediction lengths, that is {96,192,336,720}. The results highlighted in red and bold are the best results.The underlined one is the second-best result.

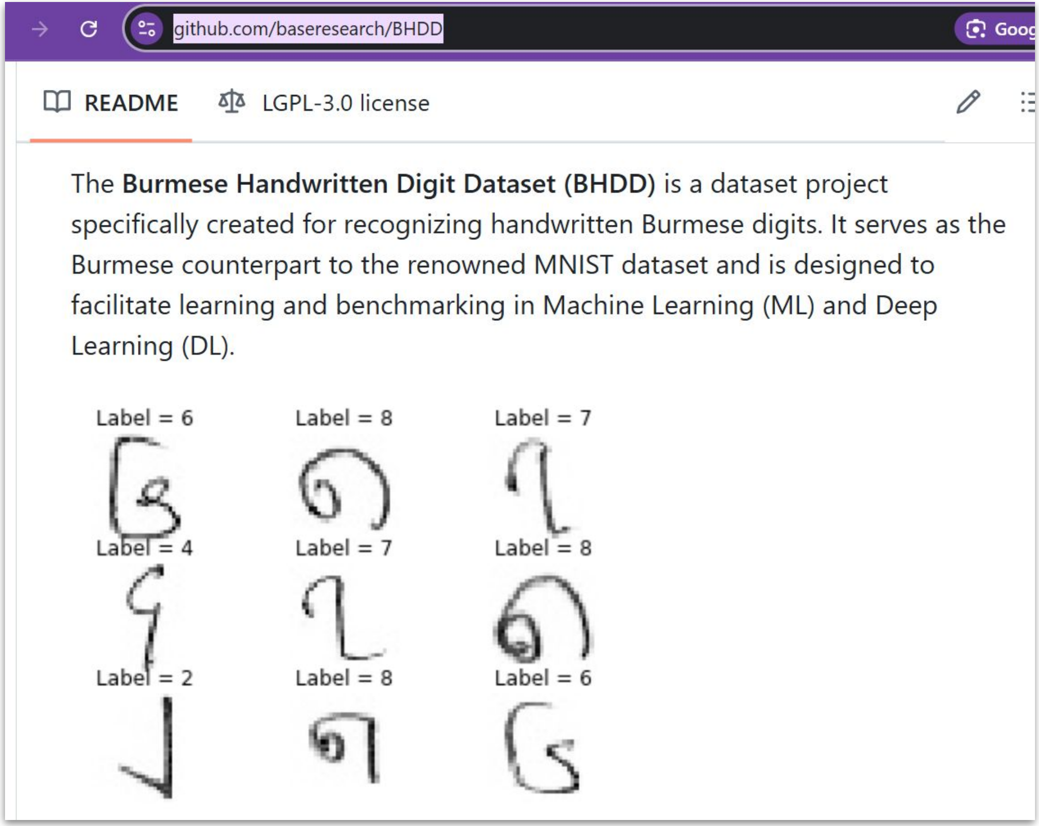
MODEL	PETNN		RNN		LSTM		GRU		TRANSFORMER	
METRIC	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTM1	**0.52**	**0.47**	1.54	0.98	1.32	0.86	1.37	0.86	<u>0.85</u>	<u>0.68</u>
ETTM2	**0.30**	**0.34**	2.80	1.36	2.39	1.18	2.26	1.10	<u>1.38</u>	<u>0.81</u>
ETTH1	**0.48**	**0.47**	1.54	0.99	1.19	0.82	1.32	0.83	<u>0.89</u>	<u>0.74</u>
ETTH2	**0.43**	**0.43**	4.87	1.64	3.09	1.35	3.43	1.49	<u>2.35</u>	<u>1.30</u>
ELECTRICITY	**0.19**	**0.29**	0.64	0.60	0.56	0.55	0.54	0.54	<u>0.32</u>	<u>0.42</u>
TRAFFIC	**0.61**	**0.33**	1.46	0.85	1.01	0.54	1.03	0.54	<u>0.68</u>	<u>0.38</u>
WEATHER	**0.27**	**0.28**	0.80	0.61	0.44	0.45	0.69	0.58	<u>0.34</u>	<u>0.47</u>
EXCHANGE	**0.47**	**0.47**	2.85	1.47	2.11	1.22	2.08	1.23	<u>1.35</u>	<u>0.93</u>

- MSE (Mean Squared Error): Lower MSE is better — values closer to 0 indicate higher prediction accuracy. It penalizes larger errors more due to squaring.
- MAE (Mean Absolute Error): Less sensitive to outliers than MSE since it doesn't square the errors. Like MSE, lower values indicate better performance.

Experimental Setting

- 2 PETNN-based classification tasks: image classification and paraphrase detection
- PETNN-based text generation
- 3 publicly available Myanmar language datasets (developed by our lab):
  - *myMNIST*
  - *myParaphrase*
  - *myPOS*

Experimental Setting (myMNIST or BHDD)



Experimental Setting (myParaphrase dataset)

```
"16027","16028","16029","တော် ပါ တယ် ကြိုးစား ပါ","တော် လိုက်တာ ကြိုးစား နော် အားမလျှော့ နဲ့","1"
"22766","22767","22768","ဘယ် လို ပဲ ဒုက္ခ တွေ အနည့်းနည့်းအဖုံဖုံ လာ ပါစေ သတ္တိ ရှိ ပါ ။","ဘယ် လို ပဲ ပျော် စရာ တွေ အမျိုးမျိုး လာ ပါစေ တည့်ငြိမ် ပါ ။","0"
"26023","26024","26025","မနက်ဖြန် နေ့ခင်း ပွဲ ဆို ရင် ကော ဘယ်လို လဲ ။","မနက်ဖြန် နေ့ခင်း ပွဲ ကြည့် ကြ မယ် ။","0"
"24002","24003","24004","မင်း ကို သူ လက်စားချေ မှာ မဟုတ်ပါ ဘူး ။","သူမ အဲ့ဒါ ကို ထပ် စဉ့်စား ခဲ့ ပါ တယ် ။","0"
```

Experimental Setting (myPOS dataset)

Table.1 Corpus information of myPOS (version 3.0)

Unit	myPOS (ver. 1.0)	Ext-1: my-zh	Ext-2: my-ko	Ext-3: ASEAN-MT my	myPOS (ver. 3.0)
Sentences	11,000	10,000	10,052	12,144	43,196
Words	239,598	103,909	106,864	114,134	564,505
Average Words/Sentence	21.78	10.17	10.64	9.40	13.07

```
တစ်/tn ကိုက်/n ကို/ppm ဝမ်/n ခုနှစ်ထောင်/tn ပါ/part ။/punc
မနှစ်/n က/ppm သူ/pron ကျွန်မ/pron ကို/ppm သင်/v ပေး/part တယ်/ppm ။/punc
ကျွန်တော့်/pron ခုံ/n သွား/v ရှာ/v မလို့/part ။/punc
အတန်း/n စ/v တာ/part ကြာ/v ပြီ/ppm လား/part ။/punc
```

Experimental Setting

- Used three different activation functions — Sigmoid, GELU, and SiLU — for the Myanmar digit classification task using the myMNIST dataset
- The goal was to compare classification performance across different gating mechanisms

PETNN (Activation Function: Sigmoid)

$$sigmoid(x) = \frac{1}{1 + e^{-x}}$$

- Classic S-shaped function that squashes inputs to (0,1)
- Prone to vanishing gradients in deep networks
- Used mainly in binary classification outputs

PETNN (Activation Function: GELU)

$$\mathrm{GELU}(x) = x * \Phi(x)$$

- Where is the standard Gaussian cumulative distribution function (CDF).

$$\Phi(x) = \frac{1}{2}\left(1 + \mathrm{erf}\left(\frac{x}{\sqrt{2}}\right)\right)$$

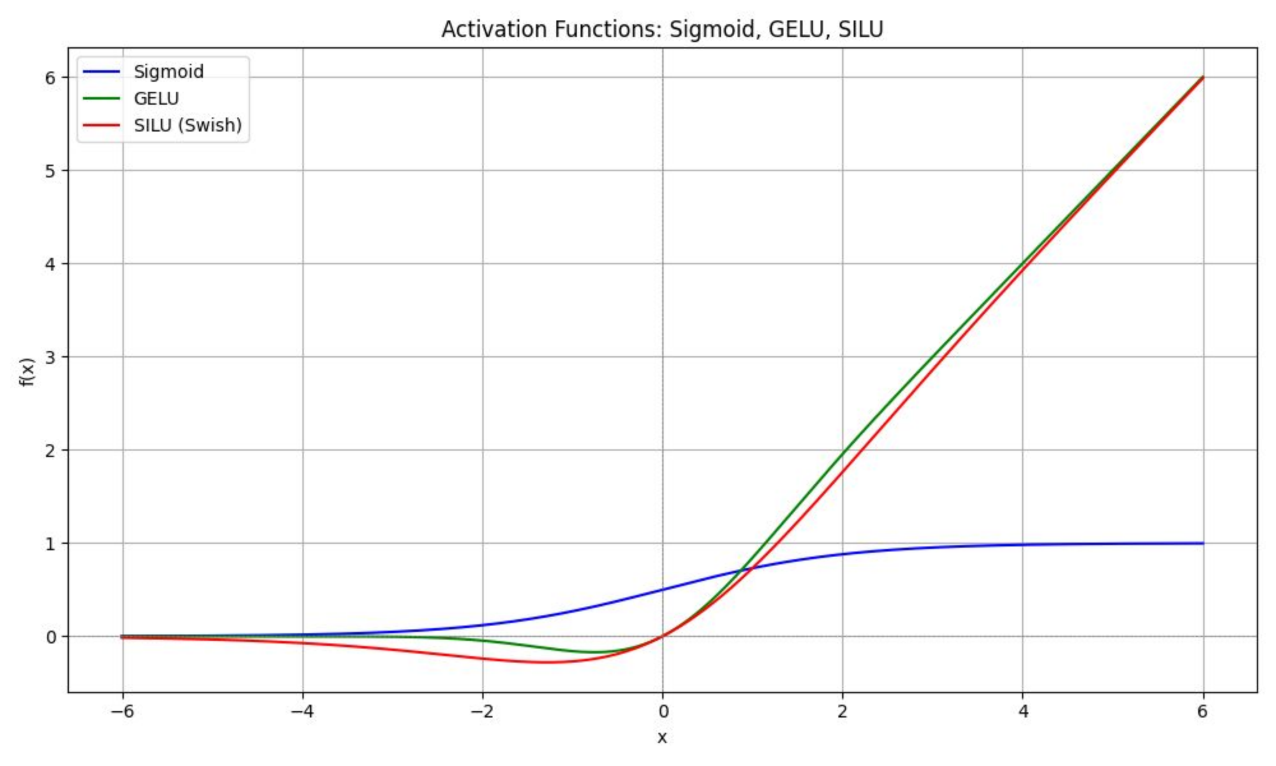
- GELU (Gaussian Error Linear Unit)
- Smooth approximation of ReLU with probabilistic interpretation
- Works well in transformers and deep networks
- Computationally more expensive than ReLU variants

PETNN (Activation Function: SiLU)

$$SiLU(x) = x * sigmoid(x)$$

- SiLU (Sigmoid-Weighted Linear Unit):
- Also called Swish (when β=1)
- Self-gating property combines benefits of ReLU and sigmoid
- Often outperforms ReLU in deep networks

PETNN (Sigmoid, GELU, SILU)



PETNN (Sigmoid, GELU, SILU)

```python
# Activation functions: Sigmoid, GELU, SILU
import numpy as np
import matplotlib.pyplot as plt
import scipy.special as sp

x = np.linspace(-6, 6, 500)

# Sigmoid
sigmoid = lambda x: 1 / (1 + np.exp(-x))

# GELU (approximation version)
gelu = lambda x: 0.5 * x * (1 + np.tanh(np.sqrt(2/np.pi) * (x + 0.044715 * x**3)))

# SILU (a.k.a. Swish)
silu = lambda x: x * sigmoid(x)

# Compute values
y_sigmoid = sigmoid(x)
y_gelu = gelu(x)
y_silu = silu(x)

# Plot
plt.figure(figsize=(10, 6))
plt.plot(x, y_sigmoid, label='Sigmoid', color='blue')
plt.plot(x, y_gelu, label='GELU', color='green')
plt.plot(x, y_silu, label='SILU (Swish)', color='red')
```

Implementation

- Referred to PETNN demo code: https://colab.research.google.com/drive/1GOnxXHP8sqMMn4AeIA2P8Cl9ozbRwXqc?usp=sharing
- Applied prompt engineering and experimented with "vibe coding" using ChatGPT, Grok, Claude
- Currently developing and debugging:
  - PETNN-Siamese model
  - PETNN-based text generation

Implementation (Let's see codes)

```
# petnn_mnist.ver3.py

import torch
import torch.nn as nn
import torch.nn.functional as F
from torchvision import datasets, transforms
from torch.utils.data import DataLoader
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
import argparse
import numpy as np
import os


class PETNNCell(nn.Module):
    def __init__(self, input_dim, hidden_dim, cell_dim, use_norm=False, dropout=0.0, gate_type='sigmoid')
        super().__init__()
        self.input_dim = input_dim
        self.hidden_dim = hidden_dim
        self.cell_dim = cell_dim
        self.concat_dim = input_dim + hidden_dim
        self.gate_type = gate_type

        print(f"[DEBUG] Initializing PETNNCell with input_dim={input_dim}, hidden_dim={hidden_dim}, "
              f"concat_dim={self.concat_dim}, gate_type={gate_type}")
                                                                                   18,9
```

SOTA Results (Classification with myMNIST)

Model	Precision	Recall	F1-Score	Accuracy
MLP	0.9810	0.9895	0.9852	0.9907
CNN	0.9955	0.9963	0.9959	0.9970
LSTM	0.9907	0.9942	0.9924	0.9951
GRU	0.9886	0.9934	0.9910	0.9937
Transformer	0.9898	0.9944	0.9921	0.9946
JEM	0.9931	0.9957	0.9944	0.9958
FastKAN	0.9844	0.9914	0.9879	0.9922
EfficientKAN	0.9841	0.9898	0.9869	0.9918
PETNN (Sigmoid)	0.9897	0.9940	0.9918	0.9943
PETNN (GELU)	0.9947	0.9963	0.9955	0.9966
PETNN (SILU)	0.9944	0.9961	0.9952	0.9964

SOTA Results (PETNN CNN Siamese for myParaphrase)

Model & Embedding	Test Type	Accuracy	Precision	Recall	F1-Score
PETNN-CNN-Siamese (w2v)	Open	0.5445	0.5815	0.5756	0.5423
PETNN-CNN-Siamese (glove)	Open	0.5516	0.5889	0.5826	0.5494
PETNN-CNN-Siamese (ft)	Open	0.5766	0.6076	0.6035	0.5758
PETNN-CNN-Siamese (w2v)	Closed	0.9800	0.9801	0.9804	0.9800
PETNN-CNN-Siamese (glove)	Closed	0.9890	0.9889	0.9892	0.9890
PETNN-CNN-Siamese (ft)	Closed	0.9550	0.9572	0.9560	0.9550

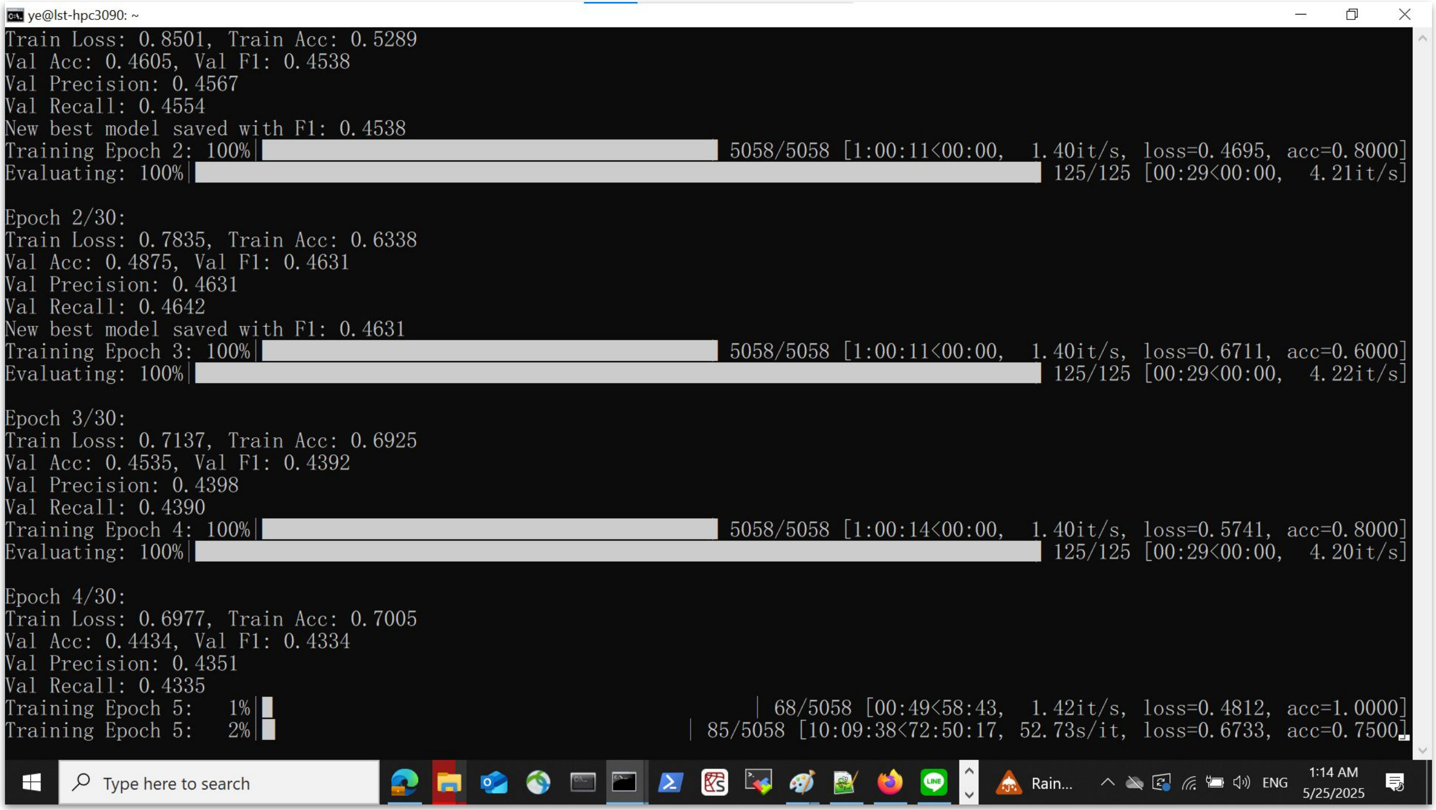
- Significant performance drop in open-test (45-58% accuracy) vs closed-test (95-99% accuracy)
- Suggests the models may be overfitting to training data distribution
- Open-test performance indicates limited generalization capability
- FastText shows best open-test performance (57.66% accuracy
- GloVe dominates in closed-test (98.90% accuracy)

SOTA Results (Our previous results for myParaphrase)

Table.1 Evaluation results on RNN-Siamese, CNN-Siamese and Transformer-Siamese with myParaphrase corpus (version 1.0)

Model	Mean-Dev-Accuracy	Last-Dev-Accuracy	Test-Acc	Training/Validation Time
bi-RNN	0.84	0.87	0.85	2m2.830s
CNN	**0.88**	**0.89**	**0.88**	**0m33.637s**
Transformer	0.81	0.82	0.81	1m38.253s

SOTA Results (Still Developing for myParaphrase)



SOTA Results (RNN text generation with myPOS)

```
Model loaded from models/burmese_rnn.pt
Example 1:
<UNK> ရာသီဥတု အေးမြ သော တောင်ထိပ် များ တောင်ကုန်း ဒေသ များ နှင့် အရှေ့မြောက်
ဘက် တွင် ဘင်္ဂလား ပင်လယ်အော် တို့ <UNK> ဝန်းရံ နေ ကြ သော်လည်း ငှက် အုပ် ကြီး
များ သည် မျောက် များ ၌ ယင်း တို့ ၏ ကိုယ်ထည် များ ကို အမှန် အားဖြင့် <UNK> ရှိ
ရယူ ထား ခြင်း ကို <UNK> ခြင်း နှင့် ချက်ချင်း ကင်းလွတ် ရေး အကျိုး အတွက် ငှာ

Example 2:
<UNK> ရာသီဥတု အဆင့်အတန်း မှ ထွက် ပြီး နှင်း များ နှင့် <UNK> <UNK> များ ကို
<UNK> လုပ်နည်း <UNK> များ ဆို စိတ်ဝင်စား လာ မှု နှင့် ယဉ်ကျေး မှု တို့ ဖြင့် ကြွယ်ဝ
ချမ်းသာ ခဲ့ ကြ သည် မှာ ၁၉ ရာစု အလယ် ပိုင်း တွင် မောင်းထုတ် နိုင် ခဲ့ ရှိ ယနေ့ အခါ ၌
ကုန် သော <UNK> များ ဖြစ် ကြ လေ သည်
```

SOTA Results (LSTM text generation with myPOS)

```
Model loaded from models/burmese_lstm.pt
Example 1:
<UNK> ရာသီဥတု အေးမြ ရှ် တောင်ထိပ် များ စွာ နှင့် ဆက်စပ် နေ သည့် ဗုဒ္ဓဟူး နေ့
နေ့လယ် ၁၂ : <UNK> နာရီ ကျော် ချိန် မွေးဖွား သူ များ အတွက် မဟာဘုတ် မှ တစ်ဆင့်
တွက်ချက် ယူ သည့် အခြား သော ဟောပြော မှု များ တွင် ပါဝင် ခဲ့ ကြ သည့် ဟု ဆို နိုင်
သည့် မှာ ရေ နှင့် သူ သည့် မနက် ဘက် ရှိ

Example 2:
<UNK> ရာသီဥတု ရှိ ခြင်း ၏ <UNK> ကို <UNK> ဟု ခေါ်ဆို ရ ခြင်း မှာ လည့်း မ
စုံလင် ပေ သည့် ဟု ပင် ပြော နိုင် သည့် မှာ <UNK> <UNK> <UNK> <UNK> <
UNK> <UNK> <UNK> <UNK> <UNK> <UNK> <UNK> <UNK> <UNK>
<UNK> <UNK> <UNK> <UNK> <UNK> <UNK> <UNK> <UNK> <UNK>
 <UNK> <UNK> <UNK> <UNK> <UNK> <UNK>
```

SOTA Results (GRU text generation with myPOS)

```
Model loaded from models/burmese_gru.pt
Example 1:
<UNK> ရာသီဥတု ရှိ တယ် ကျုန် တဲ့ နေရာ တွေ က မြို့လယ် မှာ ထား တာ တွေ သိပ် ကောင်း တယ် ဆို တာ ကျွန်တော် တို့ ဘက် မှ မ ရှိ ဘူး ဒါပေမယ့် ကျောက်ပုစွန် ကို လိုက် ကာ ပြော ပေး လိုက် ပြီး တော့ တစ် ပတ် ကို တစ် ပတ် ထား တော့ ကျင်းပ ကြ မယ် ဆို ရင် အဲဒီ လို မ လို

Example 2:
<UNK> ရာသီဥတု အေးချမ်းသာယာ ၏ အထူးသဖြင့် နွေ ရာသီ တွင် နံနက် စက် နာရီ ၅ နာရီ ၅၅ မိနစ် ခန့် တွင် နေထွက် ၏ ညနေ စက် နာရီ ၆ နာရီ ၄၃ မိနစ် တွင် နေဝင် ပြီး သည့် နောက် မှ ငှက်မ များ ဖြင့် အမြင့် ထုတ်ယူ မြင် ရ ပေ သည် ဟု ကုလသမဂ္ဂ တွင် ထပ်မံ ၏ အတည်ပြု ပြီး ကြေညာ ခြင်း မ ပြု
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Conclusion

- Reviewed the relationship between Physics and Machine Learning, including both ML for Physics and Physics for ML paradigms
- Introduced the PETNN model and its core concepts
- Demonstrated PETNN-based classification and text generation using Myanmar language datasets
- Aiming to present our state-of-the-art (SOTA) results at an international conference in the near future

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